

1. Solve the system of equations using elimination.

$$\begin{cases} 6y + 3x = -6 \\ 10x - 4y = 28 \end{cases}$$

2. Verify the solution to question 1 algebraically using substitution.

Jasper is working two different jobs over the summer. He babysits for \$9 an hour and \$15 per hour mowing lawns. Last week Jasper made a total of \$144 and worked a combined total of 14 hours. This situation can be represented by the system of equations shown where b represents the hours he babysits and l represents the amount of hours he mowed lawns.

$$\begin{cases} 9b + 15l = 144 \\ b + l = 14 \end{cases}$$

3. The system could be solved using substitution or elimination. Which method do you prefer to use?
4. Solve the system of equations using your preferred method.
5. Interpret your solution.
6. Describe the constraints on b and l in this context.

Jackson and Miriam went to a restaurant and ordered one plain pizza and two sodas. Their bill totaled \$15.95. Later that day, *five* friends went to the same restaurant. They ordered *three* plain pizzas and each person had one soda. Their bill totaled \$45.90. This situation can be represented by the system of equations shown where p represents the cost of one plain pizza and s represents the cost of a soda.

$$\begin{cases} p + 2s = 15.95 \\ 3p + 5s = 45.90 \end{cases}$$

7. The system could be solved using both substitution or elimination. Which method do you prefer to use?
8. Solve the system of equations.
9. Interpret the solution.
10. Determine the constraints on p and s in this context.